

Olympiad Test Paper — Science (Class 7)

Chapter 2: Nutrition in Animals

Subject	Science (NCERT Class 7)
Chapter	2 — Nutrition in Animals
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. All questions stay within the Class-7 chapter on Nutrition in Animals — the five steps of nutrition, the human digestive system (mouth to anus, plus liver and pancreas), feeding in Amoeba, and rumination in cows. Most questions need two to four reasoning steps; watch for traps built from common misconceptions. No diagram is printed — where a structure matters, it is described in words. Do not write on this paper — mark answers on your answer sheet.

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- The five steps of nutrition occur in a fixed order. Which sequence is correct?
(A) Ingestion → Absorption → Digestion → Assimilation → Egestion
(B) Digestion → Ingestion → Absorption → Egestion → Assimilation
(C) Ingestion → Digestion → Absorption → Assimilation → Egestion
(D) Ingestion → Digestion → Assimilation → Absorption → Egestion
 - A student says, "Once I swallow food, my body can use it straight away." The flaw in this statement is that the food must still be:
(A) ingested only
(B) digested and absorbed before cells can use it
(C) egested before it can be used (D) assimilated before it is digested
 - Which step of nutrition actually delivers nutrients **into the blood**?
(A) digestion (B) egestion
(C) assimilation (D) absorption
 - "Assimilation" is best described as the step in which:
(A) absorbed nutrients are used by cells for energy, growth, and repair
(B) complex food is broken into simple molecules
(C) food is taken into the mouth (D) undigested waste leaves the body

5. Removal of **undigested** food from the body is called egestion. This is different from excretion because excretion removes:

- (A) the same undigested food, just by another name
- (B) waste produced inside cells (such as during energy release)
- (C) water only
- (D) bacteria from the gut

6. Why can't a single molecule of starch from a roti simply pass straight into your blood the moment you swallow?

- (A) because starch is poisonous until cooked
- (B) because the starch molecule is too large until enzymes break it into simple sugar
- (C) because the blood does not touch the stomach
- (D) because starch is absorbed only in the large intestine

7. A butterfly sipping nectar and a snake swallowing a whole egg differ mainly in their step of:

- (A) absorption
- (B) assimilation
- (C) egestion
- (D) ingestion

8. An Amoeba pushes out finger-like extensions to surround a food particle. These extensions are called:

- (A) villi
- (B) cilia
- (C) flagella
- (D) pseudopodia

9. A housefly cannot chew. It pours digestive juice onto solid food and then sucks up the liquid. This shows that, in the housefly:

- (A) no digestion occurs at all
- (B) absorption happens before ingestion
- (C) some digestion begins **outside** the body before ingestion is complete
- (D) egestion happens first

10. A mosquito's mouthparts and a cow's flat-topped teeth are different because:

- (A) mosquitoes are larger than cows
- (B) the kind of food an animal eats shapes its feeding tools
- (C) all insects suck and all mammals grind, with no exceptions
- (D) cows do not need to feed

11. An adult human has 32 teeth. Which set is used mainly for **biting and cutting** food?

- (A) incisors
- (B) canines
- (C) premolars
- (D) molars

12. A carnivore such as a dog has long, pointed teeth on each side of its front teeth, well suited to **tearing** flesh. These are the:

- (A) incisors
- (B) molars
- (C) canines
- (D) premolars

13. The hardest substance in the human body, which forms the outer covering of the crown of a tooth, is:

- (A) dentine
- (B) pulp
- (C) enamel
- (D) bone

14. The central part of a tooth, which contains nerves and blood vessels, is the:

- (A) crown
- (B) enamel
- (C) root
- (D) pulp cavity

15. Tooth decay begins when bacteria act on left-over sugar and release acid that attacks the enamel. The best single reason to brush after sweets is therefore that brushing:

- (A) makes the teeth whiter only
- (B) replaces lost enamel
- (C) removes the sugar before bacteria can turn it into enamel-attacking acid
- (D) kills the tongue's taste buds

16. Counting an adult's grinding teeth: there are 8 premolars and 12 molars. The total number of grinding teeth is:

- (A) 12
- (B) 32
- (C) 8
- (D) 20

17. Saliva does more than wet the food. The enzyme in saliva begins the digestion of:

- (A) starch
- (B) fats
- (C) proteins
- (D) water

18. If you chew a plain piece of bread for a long time without swallowing, it slowly tastes sweeter. This is because:

- (A) the bread absorbs sugar from the tongue
- (B) salivary amylase breaks starch into simpler sugars
- (C) the acid in saliva caramelises the bread
- (D) bacteria add sugar to the bread

19. Three pairs of glands in the mouth release saliva. Apart from starting starch digestion, saliva also helps by:

- (A) absorbing nutrients into the blood
- (B) storing undigested waste
- (C) producing hydrochloric acid
- (D) moistening food so it can be swallowed easily

20. After you swallow, food is pushed down the food pipe by a wave-like squeezing of its muscular wall. This movement is called:

- (A) peristalsis
- (B) rumination
- (C) emulsification
- (D) absorption

21. A person lying flat on their back can still swallow water and it reaches the stomach. The reason peristalsis can move food even against gravity is that:

- (A) gravity always helps swallowing
- (B) muscles of the food pipe contract in a sequence that pushes food forward
- (C) the food pipe is a straight vertical drop
- (D) water is lighter than air

22. The food pipe (oesophagus) carries food from the mouth region to the:

- (A) stomach
- (B) liver
- (C) small intestine
- (D) large intestine

23. The stomach is a J-shaped muscular bag. Its strong walls help digestion mainly by:

- (A) absorbing all nutrients into the blood
- (B) reabsorbing water from waste
- (C) producing bile
- (D) churning the food and mixing it thoroughly with digestive juices

24. The gastric juice of the stomach contains hydrochloric acid, mucus, and a protein-digesting enzyme called:

- (A) amylase
- (B) pepsin
- (C) bile
- (D) trypsin

25. Pepsin works best in a strongly acidic setting. The hydrochloric acid in the stomach therefore has the double job of killing germs in food **and**:

- (A) cooling the food
- (B) breaking down fats into droplets
- (C) creating the acidic conditions that let pepsin act
- (D) absorbing sugar

26. The stomach is itself made largely of protein, yet pepsin and acid do not normally digest the stomach wall. The protection comes from:

- (A) bile coating the stomach
- (B) a layer of mucus lining the inner wall
- (C) the wall being made of bone
- (D) the wall having no proteins at all

27. The thick, semi-liquid paste of partly digested food that the stomach releases into the small intestine is called:

- (A) chyme
- (B) bile
- (C) cud
- (D) saliva

28. In which order does food pass through these three organs?

- (A) stomach → oesophagus → small intestine
- (B) small intestine → stomach → oesophagus
- (C) oesophagus → stomach → small intestine
- (D) oesophagus → small intestine → stomach

29. The small intestine is the **longest** part of the food canal (about 6 metres in an adult). The single biggest reason it is so long and coiled is to:

- (A) store undigested food
- (B) provide a large surface and enough time to digest food and absorb nutrients

(C) make hydrochloric acid

(D) churn food the way the stomach does

30. Two secretions empty into the first part of the small intestine to complete digestion: bile from the liver and a juice rich in enzymes from the:

(A) gall bladder

(B) stomach

(C) salivary glands

(D) pancreas

31. The inner wall of the small intestine is covered with millions of tiny finger-like projections. Their main purpose is to:

(A) grind the food

(B) greatly increase the surface area for absorption

(C) make the intestine acidic

(D) store bile

32. These finger-like projections of the small-intestine wall are called:

(A) cilia

(B) villi

(C) pseudopodia

(D) sphincters

33. Inside each villus are tiny blood vessels. This tells us that, right at the villus, the step taking place is:

(A) ingestion

(B) digestion of protein

(C) absorption of nutrients into the blood

(D) egestion

34. Suppose a disease flattened the villi so the inner wall of the small intestine became smooth. The most direct result would be:

(A) faster digestion of starch in the mouth

(B) poorer absorption of nutrients, so the person stays undernourished even while eating

(C) more bile produced

(D) stronger churning by the stomach

35. The large intestine is shorter and wider than the small intestine. Its chief job is to:

(A) digest proteins

(B) absorb digested sugars into the blood

(C) produce pepsin

(D) absorb water from the undigested residue

36. If the large intestine absorbs **too much** water from the waste, the likely result is:

(A) hard, dry stools that are difficult to egest (constipation)

(B) loose, watery stools

(C) vomiting of acid

(D) faster digestion of fats

37. The undigested, solid waste is stored for a short time and then removed through the anus. This final step is:

(A) absorption

(B) assimilation

(C) egestion

(D) ingestion

38. The largest gland in the human body, which makes bile, is the:

- (A) pancreas
- (B) salivary gland
- (C) liver
- (D) gall bladder

39. Bile is stored, between meals, in a small sac called the:

- (A) rumen
- (B) duodenum
- (C) gall bladder
- (D) pancreas

40. Bile contains **no digestive enzyme**, yet it greatly helps the digestion of fats. It does this by:

- (A) chemically breaking fat into amino acids
- (B) turning fat into starch
- (C) absorbing the fat into the blood directly
- (D) breaking large fat globules into tiny droplets so enzymes can act on them more easily

41. A patient whose gall bladder has been removed often finds **fatty** meals hardest to handle. This fits the idea that bile mainly assists the digestion of:

- (A) fats
- (B) starch
- (C) proteins
- (D) water

42. The pancreatic juice contains enzymes that act on carbohydrates, proteins, **and** fats. This makes the pancreas special because most other digestive juices:

- (A) only absorb water
- (B) contain no enzymes
- (C) are made in the mouth
- (D) act on only one or two food types

43. Cows, buffaloes, goats and deer are grouped together as **ruminants** chiefly because they:

- (A) all eat meat
- (B) digest food inside a single cell
- (C) have only one stomach chamber
- (D) swallow grass quickly and later return it to the mouth to chew again

44. The partly digested grass that a cow brings back up to chew slowly while resting is called:

- (A) cud
- (B) bile
- (C) chyme
- (D) mucus

45. A ruminant's stomach has four chambers. The largest chamber, where microbes start breaking down the grass, is the:

- (A) abomasum
- (B) rumen
- (C) reticulum
- (D) omasum

46. Grass is rich in cellulose. Cows can live on grass even though, like humans, they make no enzyme to digest cellulose. They manage because:

- (A) they cook the grass first
- (B) cellulose needs no digestion
- (C) microbes living in the rumen break the cellulose down for them

(D) their saliva contains cellulase

47. A human eats the same grass and gets almost nothing from it, but a cow thrives on it. The best single reason is that the human:

(A) lacks both the rumen microbes and the re-chewing system that break down cellulose

(B) has no stomach

(C) cannot swallow grass

(D) digests grass in the mouth

48. In which chamber of a ruminant's stomach does acid-and-enzyme digestion happen, much like in a human stomach?

(A) rumen

(B) reticulum

(C) abomasum

(D) omasum

49. Re-chewing the cud is useful because grinding the grass more finely:

(A) makes the grass taste sweeter

(B) turns cellulose into protein instantly

(C) removes the need for any stomach

(D) exposes more surface for microbes and enzymes to act on

50. Amoeba is a single-celled animal. It captures its food by surrounding it with pseudopodia. The food then ends up trapped inside a:

(A) villus

(B) gall bladder

(C) food vacuole

(D) rumen

51. Inside the Amoeba, digestive enzymes are released **into the food vacuole**. This is comparable, in a human, to enzymes acting:

(A) inside the cavity of the stomach and intestine

(B) inside the brain

(C) on the skin surface

(D) in the blood only

52. Which statement correctly compares nutrition in Amoeba and in a human?

(A) only the human carries out digestion; Amoeba does not

(B) the human takes in food but never assimilates it

(C) Amoeba absorbs food but never egests

(D) both carry out all five steps, but Amoeba does them within one cell while a human uses many organs

53. The way an Amoeba engulfs a food particle with its pseudopodia is most similar to which everyday image?

(A) a hand cupping and wrapping around a marble

(B) a straw sucking up juice

(C) a sieve straining water

(D) a knife cutting bread

54. After an Amoeba has digested and absorbed the useful part of its meal, the leftover undigested matter is:

- (A) thrown out when the vacuole reaches the cell surface (egestion)
- (B) kept forever inside the cell
- (C) turned into a new pseudopodium
- (D) passed to a neighbouring cell

55. A pseudopodium is a "temporary" structure, unlike a human's stomach. The key difference this points to is that Amoeba:

- (A) has permanent organs for each step
- (B) makes structures as needed rather than having fixed organs
- (C) cannot move at all
- (D) digests food outside its body

56. The glucose absorbed from your food is finally "burnt" inside your cells to release energy. The tiny structures in the cell where this energy release mainly happens are the:

- (A) villi
- (B) mitochondria
- (C) food vacuoles
- (D) salivary glands

57. After a meal, a growing child uses some of the absorbed amino acids to build new muscle. This use of absorbed nutrients to build and repair tissue is the step called:

- (A) ingestion
- (B) digestion
- (C) egestion
- (D) assimilation

58. A person eats well but their digestion (in the gut) works fine while their cells cannot use glucose properly. The step that is failing here is:

- (A) assimilation
- (B) digestion
- (C) absorption
- (D) ingestion

59. Trace one mouthful of rice through the body. In which organ is its **starch digestion first started** and in which is the resulting sugar **mainly absorbed**?

- (A) started in the mouth; absorbed in the small intestine
- (B) started in the stomach; absorbed in the large intestine
- (C) started in the small intestine; absorbed in the stomach
- (D) started in the liver; absorbed in the mouth

60. A meal of paneer (a protein-rich food) is eaten. Order the places where it is chiefly broken down and where its building blocks are absorbed.

- (A) protein broken down in the large intestine; absorbed in the stomach
- (B) protein broken down only in the mouth; absorbed in the large intestine
- (C) protein broken down mainly in the stomach and small intestine; building blocks absorbed in the small intestine
- (D) protein not broken down at all; absorbed whole in the mouth

Solutions — Science (Class 7), Chapter 2:

Nutrition in Animals

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	B	31	B	41	A	51	A
2	B	12	C	22	A	32	B	42	D	52	D
3	D	13	C	23	D	33	C	43	D	53	A
4	A	14	D	24	B	34	B	44	A	54	A
5	B	15	C	25	C	35	D	45	B	55	B
6	B	16	D	26	B	36	A	46	C	56	B
7	D	17	A	27	A	37	C	47	A	57	D
8	D	18	B	28	C	38	C	48	C	58	A
9	C	19	D	29	B	39	C	49	D	59	A
10	B	20	A	30	D	40	D	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The five steps run: ingestion (taking food in) → digestion (breaking it down) → absorption (passing nutrients to the blood) → assimilation (cells using them) → egestion (removing undigested waste). Only option C lists this exact order. The others scramble it — e.g. putting absorption before digestion is impossible, because nothing can be absorbed until it is first broken small enough to cross the gut wall.

2. (B) Swallowed food is large and locked in complex molecules; cells can use it only after it is *digested* into simple molecules and then *absorbed* into the blood. So the missing steps are digestion and absorption. "Ingested only" is what already happened; egestion and assimilation come later, and assimilation can never precede digestion.

3. (D) Absorption is precisely the step in which the simple nutrients pass from the small intestine into the bloodstream. Digestion only breaks food down (no blood transfer yet); assimilation is cells *using* what is already in the blood; egestion removes waste.

4. (A) Assimilation is the use of absorbed nutrients by body cells for energy, growth, and repair. Option B describes digestion, C describes ingestion, and D describes egestion —

classic step-mix-up distractors.

5. (B) Egestion removes *undigested food* that never entered the cells. Excretion is different: it removes wastes produced *inside the body's cells* (for example during the release of energy). So they are not the same process under a different name (A), nor merely water (C) or bacteria (D).

6. (B) Starch is a giant molecule; the gut wall only lets through small, simple sugar molecules. Until enzymes (starting with salivary amylase) cut the starch into simple sugars, it is too large to be absorbed. It is not poisonous (A), the blood does reach the gut wall via capillaries (C), and starch is mostly absorbed in the small intestine, not the large (D).

7. (D) A butterfly sucking nectar through a proboscis and a snake swallowing prey whole are both *taking food in* — they differ in their mode of **ingestion**. Absorption, assimilation, and egestion are later, internal steps that are broadly similar.

8. (D) Amoeba extends temporary cytoplasmic projections called **pseudopodia** ("false feet") to surround food. Villi line the human intestine; cilia and flagella are hair-like motion structures, not feeding arms in Amoeba.

9. (C) The housefly releases digestive juice onto solid food *outside* its body and then sucks up the liquefied result — so **some digestion begins outside the body** before ingestion is complete. Digestion clearly does occur (so not A), and absorption/egestion still come in their normal later order.

10. (B) Feeding tools match the food: a mosquito pierces and sucks fluids, a cow grinds tough grass with flat molars. So **the kind of food an animal eats shapes its feeding tools**. Size (A) is irrelevant, the "no exceptions" rule (C) is false (e.g. a butterfly is an insect that sucks while a grasshopper chews), and cows obviously feed (D).

11. (A) The flat, chisel-edged front teeth, the **incisors**, do the biting and cutting. Canines tear, premolars and molars grind — a different job for each shape.

12. (C) The long, pointed teeth beside the incisors are the **canines**, well shaped for tearing flesh — prominent in carnivores like dogs. Incisors cut, premolars/molars grind.

13. (C) **Enamel** is the hard outer covering of the crown and is the hardest substance in the human body. Dentine lies beneath it, pulp is the soft inner core, and bone is not part of the visible tooth surface.

14. (D) The **pulp cavity** at the centre of the tooth holds the nerves and blood vessels — which is why a deep cavity hurts. The crown is the visible top, enamel is the outer coat, and the root anchors the tooth in the jaw.

- 15. (C)** Decay starts when mouth bacteria turn left-over sugar into acid that attacks enamel. Brushing **removes the sugar before the bacteria can make that acid** — that is the real protective action. Brushing does not rebuild lost enamel (B), whiteness alone (A) is cosmetic, and it should not harm taste buds (D).
- 16. (D)** Grinding teeth = premolars + molars = $8 + 12 = 20$. (12 counts only molars; 8 counts only premolars; 32 is all the teeth, not just the grinders — three tempting miscounts.)
- 17. (A)** The enzyme in saliva (salivary amylase) begins the digestion of **starch**. Proteins are first attacked in the stomach by pepsin; fats are handled later with bile and pancreatic enzymes; water needs no digestion.
- 18. (B)** Chewing plain bread tastes sweeter over time because **salivary amylase breaks the starch into simpler sugars**, which taste sweet. The bread does not absorb sugar from the tongue (A), saliva is not hot or acidic enough to caramelize (C), and bacteria do not add sugar that fast (D).
- 19. (D)** Besides starting starch digestion, saliva **moistens the food so it can be swallowed** smoothly and rolled into a soft lump. Absorption (A) happens in the intestine, HCl (C) is made in the stomach, and waste storage (B) is the large intestine's job.
- 20. (A)** The wave-like muscular squeezing that pushes food down the food pipe is **peristalsis**. Rumination is cud-chewing in cows, emulsification is bile's action on fat, and absorption is nutrient uptake.
- 21. (B)** Swallowing works lying down because the **food-pipe muscles contract in a coordinated sequence** that actively pushes the lump forward — it does not rely on gravity (so A is false, and the pipe is not a straight vertical drop, C). Water being "light" (D) is irrelevant.
- 22. (A)** The oesophagus carries food from the throat region straight to the **stomach**. The small and large intestines come after the stomach; the liver is a gland off to the side, not on the food's swallowing path.
- 23. (D)** The stomach's muscular walls mainly **churn the food and mix it thoroughly with gastric juices**. Most absorption happens later in the small intestine (A), bile is made by the liver (C), and water reabsorption is the large intestine's role (B).
- 24. (B)** The protein-digesting enzyme in gastric juice is **pepsin**. Amylase digests starch (mouth/pancreas), bile is not an enzyme (it emulsifies fat), and trypsin is a pancreatic protein-enzyme, not a stomach one.
- 25. (C)** Besides killing germs in food, the stomach's hydrochloric acid **creates the acidic conditions that allow pepsin to work** (pepsin needs strong acid). It does not cool

food (A), break fats into droplets (B — that is bile), or absorb sugar (D).

26. (B) Even though the stomach wall is made of protein, **a layer of mucus lines the inner wall** and shields it from the acid and pepsin. The wall is not made of bone (C) and certainly contains proteins (D); bile (A) is not involved in stomach protection.

27. (A) The thick, semi-liquid paste of partly digested food released from the stomach is called **chyme**. Cud is the regurgitated grass of a ruminant, bile is the liver's fat-fluid, and saliva is the mouth's secretion.

28. (C) Following the food's path: mouth → **oesophagus** → **stomach** → **small intestine**. Option C is the only correct order; the others place the stomach or small intestine before the oesophagus, which never happens.

29. (B) The small intestine is long and coiled chiefly to **provide a large surface area and enough time to finish digestion and absorb the nutrients**. It does not store waste (A — that is the large intestine), make HCl (C — the stomach), or churn like the stomach (D).

30. (D) Into the first part of the small intestine empties bile (from the liver) and an enzyme-rich juice from the **pancreas** (pancreatic juice). The gall bladder only *stores* bile; the stomach and salivary glands act earlier in the canal.

31. (B) The finger-like projections exist to **greatly increase the surface area for absorption**, so more nutrients pass into the blood per second. They do not grind food (A), make the gut acidic (C), or store bile (D).

32. (B) These projections lining the small-intestine wall are the **villi**. Cilia are tiny beating hairs, pseudopodia belong to Amoeba, and sphincters are ring muscles that open and close passages.

33. (C) Because each villus contains tiny blood vessels (capillaries), the step happening there is **absorption of nutrients into the blood**. Ingestion is at the mouth, protein digestion happens before this in the stomach, and egestion is waste removal at the far end.

34. (B) Flattening the villi removes most of the absorbing surface, so the most direct effect is **poorer absorption — the person stays undernourished even while eating enough**. It would not speed up mouth digestion (A), increase bile (C), or strengthen stomach churning (D); those are unrelated organs.

35. (D) The large intestine's chief job is to **absorb water from the undigested residue**, turning it into solid waste. It does not digest proteins (A) or absorb sugars into the blood (B — the small intestine does that), and pepsin is a stomach enzyme (C).

- 36. (A)** If too much water is pulled out, the leftover becomes **hard, dry stools that are difficult to egest — constipation**. Too *little* water removal would cause loose stools (B); vomiting acid (C) and faster fat digestion (D) are unrelated to large-intestine water uptake.
- 37. (C)** Removing the stored solid waste through the anus is the final step, **egestion**. Absorption and assimilation happen earlier and deal with useful nutrients; ingestion is the very first step.
- 38. (C)** The largest gland in the human body, and the maker of bile, is the **liver**. The pancreas makes enzyme juice, salivary glands make saliva, and the gall bladder only stores bile (it is not a gland that makes it).
- 39. (C)** Between meals, bile is stored in the **gall bladder**, a small sac, and released when fatty food arrives. The rumen is a cow's stomach chamber, the duodenum is the start of the small intestine, and the pancreas makes a different juice.
- 40. (D)** Bile carries no enzyme, yet it helps fat digestion by **breaking large fat globules into tiny droplets** (emulsification), giving fat-digesting enzymes far more surface to act on. It does not chemically cut fat into amino acids (A — fats give fatty acids, not amino acids), turn fat to starch (B), or absorb fat directly (C).
- 41. (A)** Without a gall bladder, bile is no longer concentrated and released in a burst, so **fatty** meals are hardest to handle — confirming that bile mainly assists the digestion of **fats**. Starch (B), proteins (C), and water (D) do not depend on bile.
- 42. (D)** The pancreatic juice is special because it acts on carbohydrates, proteins, *and* fats together, whereas **most other digestive juices act on only one or two food types** (e.g. saliva on starch alone, gastric juice mainly on protein). It is full of enzymes (so not B), is not made in the mouth (C), and is for digestion, not water absorption (A).
- 43. (D)** Ruminants are grouped together because they **swallow grass quickly and later bring it back to the mouth to chew again** (chewing the cud). They are plant-eaters, not meat-eaters (A); they have a four-chambered, not single-chambered, stomach (C); and they digest in organs, not inside one cell (D refers to Amoeba).
- 44. (A)** The partly digested grass a cow returns to its mouth to chew at leisure is called **cud**. Chyme is the human stomach paste, bile is the liver's fluid, and mucus is a protective slime.
- 45. (B)** The largest of the four chambers, where microbes first attack the grass, is the **rumen**. The reticulum, omasum, and abomasum are the smaller, later chambers.
- 46. (C)** Cows cannot make a cellulose-digesting enzyme themselves; they manage because **microbes living in the rumen break the cellulose down for them**. They do not

cook grass (A), cellulose certainly needs digestion (B), and their saliva does not contain cellulase (D) — the microbes supply that ability.

47. (A) A human gets little from grass because the human **lacks both the rumen microbes and the cud re-chewing system** that together break down cellulose. The human does have a stomach (B), can swallow grass (C), and does not digest grass in the mouth (D) — none of those is the real reason.

48. (C) Acid-and-enzyme digestion, like that in a human stomach, happens in the **abomasum**, the "true stomach" chamber. The rumen and reticulum host microbial breakdown, and the omasum mainly absorbs water.

49. (D) Re-chewing the cud grinds the grass finer, which **exposes more surface for microbes and enzymes to act on**, speeding cellulose breakdown. It does not change taste (A), turn cellulose into protein (B), or remove the need for a stomach (C).

50. (C) After the pseudopodia wrap around the food, it is trapped inside a **food vacuole**, where enzymes digest it. A villus and gall bladder are human structures; the rumen is a cow's stomach chamber.

51. (A) In Amoeba the enzymes are released into the food vacuole — a closed pocket where digestion happens. The human parallel is enzymes acting **inside the cavity of the stomach and intestine** (also enclosed spaces). It is not in the brain (B), on the skin (C), or only in the blood (D).

52. (D) The correct comparison: **both Amoeba and humans carry out all five steps of nutrition, but Amoeba does them within a single cell while a human uses many specialised organs**. Amoeba does digest (so A is false), it does egest (C false), and humans do assimilate (B false).

53. (A) Pseudopodia flowing around and wrapping a food particle is most like **a hand cupping and wrapping around a marble** — surrounding and enclosing it. A straw sucking (B) is more like a butterfly's proboscis, a sieve straining (C) is like baleen filtering, and a knife cutting (D) is like incisors.

54. (A) Leftover undigested matter in Amoeba is **thrown out when the food vacuole reaches the cell surface and discharges its contents** — this is egestion in a single cell. It is not kept forever (B), recycled into a pseudopodium (C), or passed to a neighbour (D).

55. (B) A pseudopodium being temporary tells us that Amoeba **makes structures as it needs them rather than having fixed organs** — a single cell improvises. So it does **not** have permanent organs (A), it certainly can move (C), and its digestion occurs inside the cell, not outside (D).

56. (B) Glucose is "burnt" to release energy mainly inside the **mitochondria**, the cell's energy structures. Villi absorb nutrients, food vacuoles are Amoeba's digestion pockets, and salivary glands make saliva — none release the cell's energy.

57. (D) Using absorbed amino acids to build new muscle and repair tissue is the step called **assimilation**. Ingestion is taking food in, digestion is breaking it down, and egestion is waste removal.

58. (A) Digestion and absorption are working fine, but the *cells cannot use the glucose* — so the failing step is the cellular use of nutrients, i.e. **assimilation**. It is not ingestion, digestion, or absorption, all of which the question says are working.

59. (A) Starch is first attacked by salivary amylase in the **mouth**; the resulting simple sugar is **mainly absorbed in the small intestine** (through the villi). Stomach acid does not start starch digestion (B), the small intestine does not start it (C), and the liver/mouth absorption pairing (D) is nonsense.

60. (C) Protein (paneer) is broken down mainly by pepsin in the **stomach** and finished by enzymes in the **small intestine**; its building blocks (amino acids) are then **absorbed in the small intestine** through the villi. It is not handled in the mouth alone (B), nor in the large intestine (A), and protein is certainly digested rather than absorbed whole (D).